



SALAS O'BRIEN
3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Camden Farms; Sterling Residential
Location: Jefferson, GA
Description: Finley - 2A, Slab, CBonus

Date: March 22, 2024
By: TJH
House Faces: Worst NorthEast

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,567 ft ²
Volume	24,267 ft ³
Percentage Glass	12 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	11188
SEER	14
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.167	Exterior Door		R-6	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.046	Flat Ceiling		R-20 sprayed	
	R2	0.05	Vaulted		R-20 sprayed	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.29	Blinds/Draperies	SHGC 0.20
	G2					
	G3					

LOAD SUMMARY					
			Unit 1 - Whole House		
			Cooling	Heating	
Sensible Load			30044 Btu/hr	38587 Btu/hr	
Minimum Nominal Capacity			3.34 Tons	11.31 kW	
Load Density			769 SF/Ton	4.4 W/SF	
Floor Area Served			2567.0 SF	2567.0 SF	
Mechanical Ventilation			N/A CFM	N/A CFM	
System Size			3.5 Tons	15.0 kW	
Leaving Air Temperature			55.6 deg F	106.5 deg F	
Installed Capacity			733 SF/Ton	5.8 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

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[illegible]

Unit 1

Size

3.50

Tons

Airflow

1400

CFM

Heat kW

15

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer & Gar Entry	60	93	76	
Dining	55	88	71	
Hall	24	28	26	
Bed 2	88	120	104	
Bath	24	33	29	
Bed 3	97	107	102	
Breakfast	75	75	75	
Kitchen	163	52	107	
Great Rm	193	203	198	
Master Bed	166	173	170	
Master Bath	45	60	52	
Master WIC	35	45	40	
Laundry	50	18	34	
Powder	11	12	12	

[illegible]

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Level(s): First Floor
SEER 14

Typical Ceiling Height 9
Cooling Size 2.5
Heating Size 3
Best Case 2.5
DX Cooling kWh 3106
RC Heating kWh 5677
Total kWh 8783

COOLING LOAD CALCULATION					Ceiling Height		9		9		9		9		9		9		12		9		9		9		9		9									
Components					CLTD (°F)	U-Value (Btu/hr-sf-F)	Cooling Load Factor		Entire House		Foyer & Gar Entry		Dining		Hall		Bed 2		Bath		Bed 3		Breakfast		Kitchen		Great Rm		Master Bed		Master Bath		Master WIC		Laundry		Powder	
	Type	Description		Qty			Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	
Glass	G1	NorthWest	Right	(E)	16 Btu/hr-sf	38	623			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	30	492	8	131			-	-	-	-		
	G1	North		(SE)	5 Btu/hr-sf	0	0																															
	G1	NorthEast	Front	(S)	7 Btu/hr-sf	0	0																															
	G1	East		(SW)	12 Btu/hr-sf	0	0																															
	G1	SouthEast	Left	(W)	13 Btu/hr-sf	6	76									6	76																					
	G1	South		(NW)	14 Btu/hr-sf	0	0																															
	G1	SouthWest	Back	(N)	23 Btu/hr-sf	75	1761										15	352	30	704							30	704										
	G1	West		(NE)	26 Btu/hr-sf	0	0																															
	G1	Sky Light			41 Btu/hr-sf	0	0																															
Door	G1	Shaded			5 Btu/hr-sf	133	710	31.34	167	36	192				36	192									30	160			5	18								
	D1	Door		21	0.17	3.51 Btu/hr-sf	10	35			5	18																										
Above Grade Walls	D2	Door		30	0.50	15.00 Btu/hr-sf	20	300	20	300																												
	W1	NorthWest		19	0.08	1.46 Btu/hr-sf	372	544			-	-	15	22	-	-	-	-	-	-	20	29	-	-	-	-	-	138	202	110	161	88	129	-	-	-	-	
	W1	North		13	0.08	1.00 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	NorthEast		19	0.08	1.46 Btu/hr-sf	222	324	44	64	82	120	-	-	96	140	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	East		23	0.08	1.77 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	SouthEast		21	0.08	1.62 Btu/hr-sf	389	629	15	24	-	-	-	-	145	234	74	120	155	251	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	South		16	0.08	1.23 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62 Btu/hr-sf	499	806			-	-	-	-	-	-	-	97	156	98	159	-	-	-	175	283	128	207	-	-	-	-	-	-	-	-		
	W1	West		23	0.08	1.77 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Basement Walls	W1	NorthWest		19	0.08	1.46 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	North		13	0.08	1.00 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthEast		19	0.08	1.46 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East		23	0.08	1.77 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast		21	0.08	1.62 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South		16	0.08	1.23 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West		23	0.08	1.77 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W2	Grade		0	0.07	0.00 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Other	W1	Partition		10	0.08	0.77 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	Garage		30	0.08	2.31 Btu/hr-sf	193	445			20	46	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W3	Attic		15	0.05	0.80 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W3	Kneewall		15	0.05	0.80 Btu/hr-sf	269	214			21	17	30	24	-	-	-	-	-	-	-	-	-	171	136	48	38	-	-	-	-	-	-	-	-	-		
	R1	Ceiling		55	0.05	2.53 Btu/hr-sf	1870	4731	102	258	142	359	91	230	178	450	54	137	173	438	112	283	169	428	388	982	267	676	110	278	31	78	53	134	-	-		
	R2	Vaulted		55	0.05	2.75 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Crawl/Uncon.		11	0.05	0.58 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Garage		30	0.05	1.59 Btu/hr-sf	0	0			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F2	Slab		0	0.10	0.00 Btu/hr-sf	1981	0	111	-	142	-	91	-	178	-	54	-	173	-	112	-	169	-	388	-	267	-	119	-	83	-	60	-	34	-	-	
Internal	People				255 Btu/hr	4	1020			-	-	-	-	1	255			1	255					1	255	-	1	255			-	-	-	-	-	-		
	Lights (W)				3.413 Btu/hr-W	994	3393	56	191	71	242	46	157	89	304	27	92	87	297	56	191	85	290	194	662	134	457	60	205	42	143	30	102	17	58			
	Appliances (W)				3.413 Btu/hr-W	1130	3857			-	-	-	-	-	-	-	-	-	-	-	-	680	2,321	250	853			-	-	200	683							
	Infiltration	0.25	ACH		0.09 Btu/hr-cf	1709			90		115		74		144		44		140		91		137		419		216		96		67		49		28			
Vent	Volume				CF	18993		999		1278		819		1602		486		1557		1008		1521		4656		2403		1071		747		540		306				
	Area				SF	1981		111		142		91		178		54		173		112		169		388		267		119		83		60		34				
Subtotal (Btu/hr)						21177		1175		1075		461		1720		468		1889		1458		3175		3767		3248		872		682		968		220				
Safety Factor					10 %		2118																															
Sensible Cooling Load (Btu/hr)						23295																																
Nominal Tons Required (75% sens. cap.)						2.59																																
System Selection					2.5 Tons	Space CFM	1000		55		51		22		81		22		89		69		150		178		153		41		32		46		10			
HEATING LOAD CALCULATION					Ceiling Height		9		9		9		9		9		9		9		12		9		9		9		9		9		9		9			
Components					AT (°F)	U-Value (Btu/hr-sf-F)	Heating Load Factor		Entire House		Foyer & Gar		Dining		Hall		Bed 2		Bath		Bed 3		Breakfast		Kitchen		Great Rm		Master Bed		Master Bath		Master WIC		Laundry		Powder	
	Type	Description		Qty			Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	
Door	Glass	G2		54	0.29	15.7 Btu/hr-sf	252	3952	31.34	491	36	564	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	
	G2			54	0.00	0.0 Btu/hr-sf	0	0		0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0										

Typical Ceiling Height	9	Cooling Size	0.75	DX Cooling kWh	900
		Heating Size	0.75	RC Heating kWh	1505
Kneewall Height	5	Best Case	0.75	Total kWh	2405